Cross-Lingual NER Transfer from English to Danish using BERT-based Models

Anonymous ACL submission

Abstract

We propose a Named Entity Recognition (NER) project that starts with an English BERT baseline on EWT and extends to Danish via cross-lingual transfer. We compare zero-shot transfer, English-initialized Danish fine-tuning, and Danish-only training, and evaluate with strict span-level F1.

1 Topic and Current State

Transformer models (e.g., BERT, mBERT, XLM-R) are standard for NER. Performance is strong in high-resource languages such as English, while lower-resource languages often suffer from limited annotation. Prior work shows multilingual encoders can transfer entity knowledge across languages, but gains vary by language similarity, label schema, and domain.

2 New Part of Our Project

Our novelty is a controlled English→Danish transfer study grounded in the course baseline pipeline. We first build a reproducible English baseline (`bert-base-cased`) on EWT in IOB2 format. We then test multilingual transfer strategies for Danish: (i) zero-shot English-trained multilingual model on Danish, (ii) English-initialized model with Danish fine-tuning, (iii) Danish-only baseline. We also include a low-resource setting (10%, 25%, 50% Danish train data).

3 Research Question

To what extent does English supervision improve Danish NER through cross-lingual transfer compared with Danish-only fine-tuning? Subquestions:

- How strong is the English baseline?
- How effective is zero-shot transfer?
- Does English initialization improve Danish performance under limited data?

4 Method and Feasibility

We use token classification with aligned subword labels and evaluate with precision, recall, and strict span F1 (`seqeval` + course scorer). The project is feasible within the course timeline because the baseline code already runs, and transfer experiments reuse the same training/evaluation pipeline.